Interactive Gantt Chart Builder

by Fraser Davies

Research Report

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Supervised by Dr Pierre Le Bras

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Authorship Declaration

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Student name	Fraser Davies
Student ID	H00403114

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Abstract

Project management is critical for students in higher education, particularly those undertaking substantial individual projects such as honours dissertations. Despite the widespread adoption of Gantt charts as the fourth most used project management tool among professionals[6], their application in individual student dissertation contexts remains underexplored in academic literature. This dissertation investigates the development of an Interactive Gantt Chart Builder specifically designed for honours project students at Heriot-Watt University. The primary objective is to design, develop, and evaluate an interactive web-based tool that guides students through creating comprehensive project plans whilst offering supervisors visibility into student progress and planning quality.

The research methodology combines systematic requirement analysis using MoSCoW prioritisation, identifying 22 functional and 7 non-functional requirements organised as user stories, with comprehensive background research examining Gantt chart design principles, project management methodologies, and interactive visualisation techniques in educational contexts. The proposed system will be implemented using ASP.NET MVC framework with Entity Framework and SQLite, integrating D3.js for interactive visualisation capabilities that enable drag-and-drop interactions and dynamic chart rendering.

The evaluation methodology combines system testing to validate functional correctness with user-centred usability evaluation employing the System Usability Scale (SUS) and task-based assessment with honours students. This research addresses genuine challenges faced by students managing extended individual projects, including the psychological barriers to maintaining detailed project schedules and the tension between planning effort and adaptive flexibility. The findings will contribute to understanding how interactive project management tools can effectively support individual academic projects whilst developing students' project management competencies.

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1 Introduction

1.1 Background and Motivation

Project management is critical for students in higher education, particularly those undertaking substantial individual projects such as honours dissertations. Dissertations present unique challenges as they are often the largest and most complex projects students have encountered, requiring sustained effort over extended periods while juggling multiple responsibilities [41]. Effective planning, scheduling, and progress monitoring are essential skills that students must develop to successfully navigate these complex, long-term academic endeavours. However, many students struggle with translating abstract project requirements into concrete, manageable tasks with realistic timelines and dependencies, finding themselves overwhelmed by the scope and duration of dissertation work [41].

Gantt charts have emerged as one of the most widely adopted project management tools, originally developed by Henry Gantt in the early 20th century to visualise project schedules through horizontal bar representations [17, 43]. These visual tools effectively display task durations, dependencies, and timelines, making them invaluable for planning and tracking project progress [13]. In educational contexts, Gantt charts serve as pedagogical tools that help students manage their increasing responsibilities while developing time management skills. Their graphical representation visualises beginning dates, ending dates, and duration parameters, helping to track various projects over specific time periods [5].

Despite their proven utility in professional project management, traditional Gantt chart tools often present usability challenges for students who lack prior project management experience. Many existing solutions are designed for team-based projects and enterprise environments, making them overly complex for individual academic projects. Furthermore, the widespread adoption of visualisation tools in educational settings has been hindered by usability problems and the time required for instructors and students to learn and incorporate these tools into their workflows.

At Heriot-Watt University (HWU), honours project students are currently required to submit project management plans as part of their dissertation deliverables. However, the existing project management system (projects.hw.ac.uk)lacks integrated tools that guide students through the process of creating well-structured Gantt charts with appropriate task hierarchies, dependencies, and milestones. This gap in the current system motivated the development of an interactive Gantt chart builder specifically tailored to the needs of honours project students and their supervisors.

1.2 Aim and Objectives

The primary aim of this project is to design, develop, and evaluate an interactive Gantt chart builder that will be integrated into the existing HWU honours project management system.

This tool will provide structured guidance to help students create comprehensive project plans while offering supervisors visibility into student progress and planning quality.

The specific objectives of this project are:

(1) To conduct comprehensive background research on Gantt chart design principles, project management methodologies, and interactive visualisation techniques relevant to educational contexts
(2) To gather and analyse requirements from key stakeholders to ensure the tool meets user needs
(3) To design an intuitive interface that guides students through defining milestones, tasks, hierarchies, and dependencies without requiring prior project management expertise
(4) To develop an interactive Gantt chart visualisation using D3.js [15, 46], ensuring seamless integration with HWU's existing technological stack
(5) To implement functionality allowing students and supervisors to explore project timelines, track progress, and re-evaluate goals as projects evolve
(6) To provide export capabilities enabling students to include Gantt charts and associated data in their project documentation
(7) To conduct usability evaluation through user studies with students and supervisors to assess the tool's effectiveness

1.3 Organisation

The organisation of this report is as follows:

- Section 2: Presents the background research, discussing related work found in the technical literature and its relevance to the project.
- Section 3: Describes the research methodology, including requirements gathering, system design approach, and evaluation methods.
- Section 4: Examines the technical and practical feasibility of the proposed system, including technology selection and integration with HWU's infrastructure.
- Section 5: Summarises the findings, discusses the implications of the work, and outlines potential directions for future research and development.

2 Background Research

2.1 Gantt Charts and Project Management

The Gantt chart traces its origins to the early twentieth century and the scientific management movement. American mechanical engineer and management consultant Henry Gantt first described a version of his charts in 1903, published alongside Frederick W. Taylor's influential work on shop management, though scholars debate whether their widespread adoption began during the First World War [49]. Conceived as production planning tools, Gantt charts monitored worker performance and machine utilisation, embodying the principles of scientific management through rational analysis, clear task durations, and systematic improvement [49]. Crucially, they addressed the factory-wide machine loading problem, without which Taylor's planning office would have been unworkable [49]. While project management applications of Gantt charts were initially limited, their paper-based forms declined in the 1950s and 1960s. They regained popularity with computerised methods in the mid-1960s and were further revitalised by microcomputer project software in the 1980s. Today, Gantt charts have evolved into flexible, software-based tools central to modern project management [49].

Building upon Henry Gantt's original design, modern Gantt charts have evolved to incorporate sophisticated features that support complex project planning. A comprehensive Gantt chart integrates multiple elements: [13, 43]

- A vertical list of tasks or activities on the left side.
- A horizontal timeline indicating project duration (days, weeks, or months)
- Graphical bars representing individual task durations and their temporal placement.
- Lines or arrows indicating dependencies between tasks.
- Labels showing team members or assignees.
- Special markers (typically diamonds) that denote milestones or important project achievements.

Task dependencies are fundamental to effective Gantt chart usage, defining the relationships between tasks and dictating the order in which activities must occur. Project management identifies four primary dependency types [36, 42]. The Finish-to-Start (FS) dependency, where a successor task cannot commence until its predecessor concludes, represents the most common relationship type [33]. Start-to-Start (SS) dependencies allow a task to begin only after another task has also began, enabling parallel workflows that share a start point. Finish-to-Finish (FF) relationships constrain task completion, requiring one task to finish before another can conclude. Finally, the Start-to-Finish (SF) dependency, the least common type, denotes that a task cannot finish until another task begins, typically appearing in handover or transition scenarios.

Beyond dependencies, Gantt charts visualise the critical path the longest sequence of dependent tasks that determines the project's minimum completion time. This enables project managers to distinguish between tasks that are critical to meeting deadlines and those that possess scheduling flexibility [4]. This distinction proves particularly valuable for resource allocation and risk management decisions.

Gantt charts occupy a distinct position within the broader landscape of project management methodologies. Unlike Kanban boards, which emphasise workflow states and work-in-progress limits while focusing on the volume of work completed, in progress, and not yet started, Gantt charts provide explicit temporal representations of task scheduling and dependencies displayed on a visual timeline [38]. Similarly, while Program Evaluation and Review Technique (PERT) charts focus on identifying critical paths through network diagrams using nodes and arrows to represent task dependencies, Gantt charts offer a more intuitive timeline-based bar chart visualisation that facilitates stakeholder communication and progress tracking [27, 37].

Within project management practice, Gantt charts serve multiple critical functions that extend beyond simple schedule visualisation. As planning tools, they enable project managers to structure complex projects by breaking them into manageable tasks with defined durations and sequences [47]. Their visual representation of timelines, task dependencies, and resource allocation facilitates effective coordination and communication among project stakeholders [47], ensuring all team members maintain a shared understanding of project progress and responsibilities [47]. Furthermore, Gantt charts function as monitoring instruments through features such as baseline comparisons and progress tracking, allowing project managers to identify deviations early and make informed decisions about resource reallocation or timeline adjustments [47]. This combination of planning, communication, and monitoring capabilities has established Gantt charts as one of the most frequently used project management tools, ranking fourth among 70 tools and techniques in a survey of 750 project managers [6, 47].

The application of Gantt charts and project management tools in educational contexts, particularly for individual student projects, remains an underexplored area in academic literature. While Bednjanec and Tretinjak demonstrate educators' use of Gantt charts for planning school activities and curriculum development [5], research on project management tools in higher education focuses predominantly on institutional IT departments [24] and collaborative group work [19]. This focus on institutional and collaborative applications leaves a gap in understanding how individual students might effectively use Gantt charts for dissertation management. Existing project management research centres on professional contexts [6], where tools are designed for team-based environments with different requirements than individual academic projects.

2.2 Gantt Charts in Educational Contexts

Within educational contexts, Gantt charts serve multiple pedagogical functions. For educators, Gantt charts provide a structured approach to planning school activities, curriculum development, and lesson planning, with their graphical representation of beginning dates, ending dates, and duration parameters proving particularly effective for tracking various educational activities over specific time periods [5]. Beyond their use by educators, Gantt charts offer valuable support for students undertaking complex academic projects such as dissertations. The ability to visualise project timelines and decompose large, complex undertakings into manageable components with clear deadlines proves particularly valuable when managing extended research projects [41]. This visual representation of project structure enables both students and educators to conceptualise and monitor multiple activities across extended timeframes [5].

However, students often encounter challenges when applying Gantt charts to individual academic projects, particularly concerning the maintenance and updating of project timelines throughout extended research periods. Research with PhD students reveals that while Gantt charts can be beneficial for planning, the unpredictable nature of academic research makes it difficult to maintain detailed charts over multi-year projects, with students reporting frustration when plans deviate from initial timelines [1]. This psychological barrier to updating charts is compounded by the investment required: as one PhD student observes, "if one invest a lot of time and energy in creating a detailed Gantt chart, then one is not gonna be willing to adjust this chart when things didn't work out as planned" [1]. This reluctance to revise detailed plans creates a tension between the effort invested in initial planning and the flexibility required for successful academic project management.

Traditional Gantt chart tools required frequent manual updates to remain accurate, which created a significant maintenance burden. Although modern software reduces this through automation, regular updates are still necessary, and increasing project complexity, particularly in large projects like a dissertation, makes Gantt charts challenging to manage [31]. Modern interactive Gantt chart platforms address many of these challenges through features such as drag-and-drop task scheduling, automatic dependency adjustments, and real-time collaborative updates, which significantly reduce the effort required to maintain project schedules and enable more flexible adaptation to changing project requirements [44]. These interactive capabilities prove particularly valuable for complex projects where tasks and timelines must be frequently adjusted, as they allow users to modify schedules efficiently without the cumbersome manual recalculation required by static chart implementations [44].

The distinction between individual and collaborative project contexts significantly influences the application of project management tools like Gantt charts. In team-based academic projects, collaborative project management platforms provide mechanisms for distributed task assignment, shared responsibility, and collective accountability, enabling group members to coordinate schedules, track individual contributions, and maintain visibility of project progress [19]. These collaborative features address common challenges in group work, including coordination costs, workload equity concerns, and the need for clear role definition among team members [19]. However, individual academic projects such as dissertations present fundamentally different challenges. Unlike team projects where collective objectives and shared evaluation create mutual accountability, dissertation students bear sole responsibility for project completion and are evaluated individually on their work [39]. Research into academic project management reveals that while research involving multiple students may appear collaborative, the structure actually consists of professor-student duos where each student manages their own sub-project as a junior partner rather than as part of an integrated team [39]. This absence of collective objectives and team-based evaluation means individual dissertation students must rely entirely on autonomous use of project management tools without the distributed workload or accountability structures present in genuine team contexts [6]. The challenges become especially pronounced when individual students encounter the psychological barriers associated with maintaining detailed project schedules throughout extended research periods, as the sole responsibility for both planning and plan maintenance creates tension between the effort invested in initial timeline development and the flexibility required for adaptive project management [1].

2.3 Interactive Data Visualisation

Effective data visualisation relies on fundamental principles that ensure clarity and comprehension. Key design principles include knowing the audience, maintaining simplicity, choosing appropriate chart types, using colour strategically whilst maintaining accessibility, and providing proper context through titles, labels and scales [25]. Interactive visualisation differs fundamentally from static visualisation in its approach to user engagement: whilst static visualisation employs an author-driven approach where a single data story is communicated, interactive visualisation adopts a reader-driven approach that enables users to extract multiple views and explore data stories dynamically through inputs such as mouse clicks or keyboard commands [30]. Interactive graphics can provide crucial information that cannot be obtained from static graphics, making them valuable tools for promoting transparency and enabling additional exploration of datasets [48]. The benefits of interactivity in educational contexts are substantial, as:

- Interactive visualisations transform data into engaging learning experiences that enhance comprehension, improve retention, and enable personalised exploration of complex concepts [23].
- User interaction paradigms for visualisation tools encompass several distinct approaches: drag-and-drop interactions enable intuitive grouping, reordering and moving of objects through direct manipulation, providing clear signifiers and immediate visual feedback throughout the interaction [28].
- Form-based interfaces present users with structured options through menus and input fields that guide interaction, allowing menu selection by keying in associated numbers or letters whilst graphical menus employ pointing devices [11].
- Direct manipulation interfaces let users interact with on-screen objects through physical, step-by-step actions that provide instant visual feedback. For example, users can resize a graphic or drag an item to a new position [40].

These interaction paradigms collectively support varied user needs and task requirements, with direct manipulation particularly valued for its ability to exploit perceptual and motor skills whilst reducing reliance on linguistic comprehension [40].

3 Methodology

3.1 Stakeholders and Use Cases

The Interactive Gantt Chart Builder system serves distinct stakeholder groups within the academic environment, each with specific needs and interactions with the system. Understanding these stakeholders and their use cases is essential for establishing clear system boundaries and functional requirements.

The primary stakeholders for Interactive Gantt Chart Builder include:

- **Students:** The primary users who create and manage dissertation project timelines, track progress, and maintain schedules throughout the academic year. The system must accommodate varying levels of experience with project management tools.
- **Supervisors:** Academic staff who monitor student progress through view-only access to Gantt charts. This enables supervisors to assess progress, identify issues, and provide timely guidance across multiple students without administrative burden.
- **Project System Developers:** The technical team responsible for system maintenance and enhancement. The Gantt chart builder must integrate seamlessly with existing HWU infrastructure while maintaining data consistency and extensibility.
- **HWU Project Database:** The external system storing student project information, supervisor assignments, and submission records. Bidirectional data synchronisation ensures consistency across the project management ecosystem.

Use cases are availible in the appendix Section A. Figure 2 in the appendix illustrates the primary use cases for the Interactive Gantt Chart Builder system, organised around the key functional areas identified in the requirements analysis. The use case diagram employs standard UML notation to represent actors, use cases, and their relationships, providing a visual representation of system functionality from the users' perspective.

- **Create Project** represents the foundational use case where students establish their dissertation project with essential metadata including title, description, and dates. The diagram shows this extends to editing project details as understanding evolves.
- **Create Tasks** allows students to break down their dissertation into manageable components with name, description, dates, and duration. The diagram shows this extends to editing and deleting tasks, with guided input to assist inexperienced users.
- **Create Milestone** enables students to mark significant deliverables and deadlines.
- **Create Task Dependency** enables students to define the task relationships discussed previously [6].
- **Update Task Status** provides progress tracking, allowing students to mark tasks as not started, in progress, or completed.

- **View Project Overview** offers students a comprehensive summary of tasks, milestones, and progress. This viewing capability extends from project creation and connects to Gantt chart visualisation, providing quick status checks for supervision meetings.
- **View Schedule Status** enables students to assess whether they are ahead, on track, or behind schedule based on completion status and dates. The system provides visual indicators highlighting schedule health, helping students develop time management skills through explicit feedback.
- **Add Supervisor Comments** enables supervisors to provide feedback directly within the planning context.
- **Save Data to Project** ensures all project information persists to the HWU database.

The relationships employ standard UML conventions: *extend* indicates optional functionality building upon a base use case, while *include* represents mandatory sub-processes [3].

3.2 Requirement Analysis

The requirement analysis for the Interactive Gantt Chart Builder was conducted through examination of existing literature on project management tools in educational contexts, consultation of stakeholder needs identified through preliminary discussions with the current system developers, and analysis of the challenges students face when managing individual dissertation projects. This process identified five core functional requirements that form the foundation of the proposed system.

The requirements are categorised using the MoSCoW method into Must have (M), Should have (S), Could have (C), and Won't have (W) [35]. To label them they are indicated FR[#] to represent Functional Requirement number [#] and NFR[#] to represent Non-Functional Requirement number [#]. Each requirement has a User Story[10] format description to clarify the requirements purpose and context.

- **FR1 - M - Project Data Creation and Management**
  As a student, I must be able to create and store my project information (title, description, start date, end date) so that I have a foundation for my Gantt chart.
- **FR2 - M - Task Creation and Editing**
  As a student, I must be able to create, edit, and delete tasks with names, descriptions, start dates, and end dates so that I can break down my dissertation into manageable components.
- **FR3 - M - Task Dependencies**
  As a student, I must be able to define dependencies between tasks (Finish-to-Start relationships) so that my project plan reflects the logical sequence of work.
- **FR4 - M - Gantt Chart Visualisation**

As a student, I must be able to view my tasks and milestones as a visual Gantt chart using D3.js so that I can understand my project timeline at a glance.

- **FR5 - M - Data Saving**
  As a student, I must be able to save my project data to a database (SQLite) so that my work is not lost between sessions.
- **FR6 - M - Milestone Creation**
  As a student, I must be able to create milestones with specific dates so that I can mark key deliverables and deadlines in my dissertation timeline.
- **FR7 - S - Guided Task Creation Interface**
  As a student, I should receive contextual help and validation when creating tasks so that I create well-structured, realistic project plans.
- **FR8 - S - Interactive Timeline Adjustment**
  As a student, I should be able to drag and drop tasks on the Gantt chart to adjust dates and durations so that I can quickly adapt my plan as circumstances change.
- **FR9 - S - Automatic Dependency Adjustment**
  As a student, when I modify a task's dates, the system should automatically update dependent tasks so that I maintain a logically consistent schedule.
- **FR10 - S - Progress Tracking and Status Updates**
  As a student, I should be able to mark tasks as not started, in progress, or completed so that I can track my actual progress against my plan.
- **FR11 - S - Progress Indicator and Timeline Comparison**
  As a student, the system should tell me if I am ahead of schedule, on track, or behind schedule so that I can take corrective action when needed.
- **FR12 - S - Gantt Chart Export (PNG/JPEG)**
  As a student, I should be able to export my Gantt chart as an image (PNG/JPEG) so that I can include it in my dissertation appendix and supervision meeting documents.
- **FR13 - S - Supervisor View-Only Access**
  As a supervisor, I should be able to view my students' Gantt charts in read-only mode so that I can monitor their progress and provide guidance.
- **FR14 - S - Task Hierarchy**
  As a student, I should be able to create subtasks under parent tasks so that I can organise complex phases of work hierarchically.
- **FR15 - S - Impossible Dependencies**
  As a student, the system should prevent me from creating logically impossible schedules.
- **FR16 - S - Timeline Zoom and Pan Controls**
  As a student, I should be able to zoom in/out and pan across the timeline so that I can view different levels of detail from my entire project to individual weeks.
- **FR17 - S - Hover Tooltips with Task Details**
  As a student, when I hover over a task bar, I should see detailed information (task name, dates, duration, dependencies) so that I can quickly access task details without navigating away.
- **FR18 - C - Kanban Board View**

As a student, I could have an alternative Kanban board view of my tasks so that I can visualise my work in a different format that shows current work in progress.

- **FR19 - C - Task Categories/Tags**
  As a student, I could assign categories or tags to tasks (e.g., "Research", "Writing", "Data Collection") so that I can filter and organise my work by type.
- **FR20 - C - Multiple Dependency Types**
  As a student, I could define different dependency types (Start-to-Start, Finish-to-Finish, Start-to-Finish) beyond the basic Finish-to-Start so that I can model more complex task relationships.
- **FR21 - C - Critical Path Highlighting [6]**
  As a student, I could view the critical path highlighted on my Gantt chart so that I can identify which tasks have no scheduling flexibility [6].
- **FR22 - C - Supervisor Comment and Feedback System**
  As a supervisor, I could add comments and feedback on specific tasks or the overall project plan so that I can provide guidance directly within the tool.
- **NFR1 - M - Web Browser Compatibility**
  As a student, I must be able to access and use the Gantt chart builder on any modern web browser (Chrome, Firefox, Safari, Edge) without installing additional plugins or software so that I can work on my project plan from any computer.
- **NFR2 - M - Data Security and GDPR Compliance**
  As a student, the system must store all my project data securely and be GDPR-compliant.
- **NFR3 - S - Response Time Performance**
  As a student, the system should render my Gantt chart and respond to my interactions within 2 seconds (for projects with up to 100 tasks) so that I can work efficiently without frustrating delays that discourage regular use.
- **NFR4 - S - Intuitive User Interface**
  As a student, the system should provide a clear, intuitive interface that I can use effectively without extensive training so that I can focus on planning my dissertation rather than learning how to use the tool.
- **NFR5 - S - Mobile-Friendly Responsive Design**
  As a student, the system should adapt its interface for my tablet and mobile devices so that I can view and ideally edit my project plan while away from my computer.
- **NFR6 - C - Accessibility Compliance**
  As a student with disabilities, the system could meet WCAG 2.1 Level AA accessibility standards with keyboard navigation, screen reader support, and sufficient color contrast so that I can use the tool effectively regardless of my accessibility needs.
- **NFR7 - C - Scalability for Multiple Concurrent Users**
  As a student, the system could support at least 200 concurrent users without performance degradation so that I can access my Gantt chart reliably even during peak usage periods like the start of semester or project deadlines

3.3 Evaluation Method

System testing will be employed to validate that the implemented system satisfies its specified functional requirements (FR1-FR22) and non-functional requirements (NFR1-NFR7). System testing is appropriate for this project as it evaluates the complete, integrated system against its requirements specification [16], which is essential for verifying that the web-based application functions correctly as a whole rather than merely as isolated components.

The system testing approach will utilise a requirements traceability matrix that maps each functional requirement to specific test cases, ensuring comprehensive coverage. This approach builds on the concept of requirements traceability as discussed by Gotel and Finkelstein [18]. Test cases will be designed to verify both positive scenarios (expected use cases) and negative scenarios (error handling and validation), following the principles outlined by Myers [32]. Particular attention will be given to the interactive features that distinguish this system from static project management tools.

Critical areas for system testing include:

- **Data persistence and integrity**: Verifying that project data, tasks, dependencies, and milestones are correctly saved to and retrieved from the SQLite database (FR5).
- **Dependency management**: Validating that task dependencies function correctly, including automatic adjustment of dependent tasks when parent tasks are modified (FR3, FR9) and prevention of logically impossible schedules (FR15).
- **Interactive visualisation**: Confirming that the D3.js-based Gantt chart accurately represents task data and responds correctly to user interactions such as drag-and-drop timeline adjustments (FR4, FR8).
- **Progress tracking functionality**: Testing the status update mechanisms and schedule comparison features to ensure accurate tracking of project progress (FR10, FR11).

Performance testing will be conducted to validate NFR3, measuring system response times under various conditions including different numbers of tasks (up to 100) and multiple concurrent users. Browser compatibility testing across Chrome, Firefox, Safari, and Edge will verify NFR1, while responsive design testing on various screen sizes will assess NFR5. Accessibility evaluation will be conducted against WCAG 2.1 Level AA standards, ensuring compliance with requirements such as keyboard navigation, screen reader support, and sufficient color contrast. In line with these standards, Garrido et al. [14] demonstrate how client-side refactoring can enhance usability for visually impaired users, underscoring the importance of adaptable interfaces in meeting accessibility needs. Together, these evaluations ensure that the system is tested not only for performance and compatibility but also for adaptability and accessibility, providing a comprehensive validation of the non-functional requirements.

Usability evaluation is essential for assessing whether the system achieves its core objective of helping students create comprehensive project plans through an intuitive, easy-to-use interface (NFR4) [45]. The usability study will recruit participants from the target user population-honours students at Heriot-Watt University who are planning or currently undertaking dissertation projects.

The usability evaluation will employ a task-based assessment methodology where participants complete a series of realistic scenarios using the Interactive Gantt Chart Builder [22]. These scenarios will be designed to exercise key functional requirements including:

- Creating a new project and defining initial project parameters (FR1)
- Adding tasks with dependencies and milestones (FR2, FR3, FR6)
- Using interactive features to adjust the timeline (FR8)
- Tracking progress and interpreting schedule status (FR10, FR11)
- Exporting the Gantt chart for documentation purposes (FR12)

Quantitative usability metrics will be collected during these tasks, including:

- Task completion rates and time-on-task measurements
- Number of errors encountered during task completion
- System Usability Scale (SUS) scores to provide standardised usability assessment [8]

Qualitative feedback will be gathered through post-task questionnaires and semi-structured interviews, focusing on participants' perceptions of the system's usefulness for dissertation planning, ease of learning (addressing NFR4), and the value of interactive features compared to static planning approaches. The questionnaire will be designed to gather specific feedback on:

- The clarity and helpfulness of the guided task creation interface (FR7)
- The effectiveness of the visual Gantt chart representation (FR4)
- The usefulness of interactive timeline adjustment compared to form-based editing (FR8)
- Overall satisfaction with the tool for managing dissertation projects

This combination of requirements-based system testing and user-centered usability evaluation provides comprehensive assessment of both technical correctness and practical utility, ensuring that the system not only functions as specified but also effectively supports students in planning their honours projects [9].

4 Feasibility

4.1 Stakeholder Meetings

The technical infrastructure and integration requirements were established through consultation with Kayvan Karim, a key stakeholder in the university's honours project management system. Karim provided guidance on technical constraints and integration approach while leaving the specific design and implementation details open-ended, with no prescribed visual requirements or timeline expectations.

While the existing honours project system architecture uses CHTML, there are no restrictions on using standard HTML and modern web technologies for the Gantt chart component. Integration will occur through a dedicated access point on the existing projects page, positioned above the ethics form section. The system will inherit existing authentication infrastructure using local email-based login.

4.2 Tools

The system will be implemented using ASP.NET MVC framework with Entity Framework for database operations. SQLite will serve as the database solution. The frontend will use standard HTML and JavaScript, with D3.js providing interactive visualisation capabilities for drag-and-drop interactions and dynamic chart rendering.

4.3 Interface Mock-ups

Preliminary interface mock-ups were created using D3.js to visualise the layout and interaction flow of the Gantt chart builder, drawing inspiration from examples in Data Visualisation Resources and The Chartmaker Directory [2, 29].

The mock-up matches the current HWU honours project system's visual style, ensuring consistency. It illustrates the user workflow from project setup through task definition to timeline interaction, demonstrating how visual elements such as milestones (amber diamonds) and tasks (coloured horizontal bars) may appear in the final implementation.

4.4 Limitations and Risks

The primary limitations of this project relate to scope and evaluation constraints. The system will focus exclusively on individual honours project management, with collaborative features and advanced dependency types (Start-to-Start, Finish-to-Finish) deferred to future development. The evaluation will be conducted with a limited sample of honours students within a single academic cycle, which may not capture long-term usability patterns or the full range of dissertation project types.

Technical risks are minimal given the use of established technologies (ASP.NET MVC, Entity Framework, D3.js) and straightforward integration with existing HWU infrastructure. The main implementation risk involves ensuring the interactive visualisation performs adequately with complex project schedules containing numerous tasks and dependencies, though preliminary testing with D3.js mock-ups suggests this is manageable within the specified performance requirements (NFR3).

5 Conclusion

This report establishes the foundation for developing an Interactive Gantt Chart Builder specifically designed for honours project students at Heriot-Watt University. The background research demonstrates that whilst Gantt charts rank as the fourth most widely used project management tool among professionals [6], their application in individual student dissertation contexts remains underexplored in academic literature.

The proposed system addresses genuine challenges faced by students managing substantial individual projects over extended periods [41]. Through systematic requirement analysis using MoSCoW prioritisation [35], this research has identified 22 functional and 7 non-functional requirements organised as user stories [10]. These encompass core capabilities including project data management, task dependencies, interactive D3.js-based visualisation [15], progress tracking, and supervisor access.

The feasibility analysis confirms technical viability through ASP.NET MVC framework with Entity Framework and SQLite, ensuring seamless integration with HWU's existing infrastructure. The evaluation methodology combines system testing [16] to validate functional correctness with user-centred usability evaluation [8] to assess practical utility, ensuring the system effectively supports students in planning and managing their honours projects. Collectively, these strategies establish a robust framework for achieving the seven objectives defined in Section 1.2.

Future work will focus on system implementation, comprehensive testing, and rigorous usability evaluation with honours students and supervisors to assess the tool's effectiveness in supporting successful dissertation project completion.

6 Implementation and Contributions

This section documents the implementation of the Interactive Gantt Chart Builder, describing how the system was built, the key technical decisions made during development, and how the final implementation addresses the requirements defined in Section 3.2. The system was implemented as a fully independent, locally-hosted web application using Node.js with Express, SQLite, and D3.js, diverging from the ASP.NET MVC stack proposed during the feasibility phase. This decision was taken following further technical consultation, which confirmed that a Node.js/Express architecture offered a more straightforward development workflow given the project's standalone scope and the absence of a requirement to directly integrate with the existing HWU system's backend during this phase.

6.1 System Architecture

The application follows a client-server architecture, with an Express server handling all REST API endpoints and a SQLite database managed via the `sqlite3` library providing persistent storage. The frontend consists of static HTML, CSS, and JavaScript files served directly by Express. Authentication uses a local email-based login system without passwords, with sessions managed via `localStorage` on the client side. The system is divided into two main modules: a login and dashboard module (`src/login-system`) and the Gantt chart builder module (`src/gantt-builder`), each with their own public assets and server-side routes. This separation keeps concerns distinct and allows each module to be reasoned about independently.

6.2 Database Design and Data Persistence

The database schema, defined in `src/login-system/database/setup.js`, implements the core entities identified during requirements analysis. The schema comprises eight tables: `students`, `student_projects`, `tasks`, `dependencies`, `subtasks`, `supervisors`, `supervisor_projects`, and `comments`. This directly addresses the data persistence requirement **FR5** and underpins all other data-related requirements.

The `students` table records each user's email address and display name, serving as the authentication source for the email-based login system. The `student_projects` table stores the title, description, start date, and end date for each project, satisfying **FR1**. Projects are associated with students via a foreign key on `student_id`, supporting a many-to-many relationship through this junction table.

The `tasks` table stores the task name, description, start and end dates, a `progress_percentage` field enforced by a `CHECK` constraint to the range `0–100`, an `is_milestone` integer flag, a `colour` field for tag-derived styling, a `display_order` field, and a `tag` field for category assignment. A `parent_task_id` column provides a self-referencing foreign key back to `tasks(task_id)`, supporting hierarchical grouping. This single table design supports **FR2**, **FR6**, **FR10**, **FR14**, and **FR19**. The `dependencies` table stores predecessor and successor task IDs alongside a `dependency_type` field (defaulting to `'finish-to-start'`), supporting **FR3**. A separate `subtasks` table holds fine-grained subtask data — name, dates, progress, and a `parent_task_id` referencing the parent task — keeping subtask records distinct from top-level tasks. An excerpt of the core table definitions is shown below:

```

CREATE TABLE IF NOT EXISTS tasks (
    task_id              INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
    project_id          INTEGER NOT NULL,
    task_name           TEXT    NOT NULL,
    start_date          TEXT,
    end_date            TEXT,
    progress_percentage INTEGER DEFAULT 0
        CHECK(progress_percentage >= 0 AND progress_percentage <= 100),
    is_milestone        INTEGER DEFAULT 0,
    parent_task_id      INTEGER,
    display_order       INTEGER DEFAULT 0,
    colour              TEXT    DEFAULT '#4a90d9',
    tag                 TEXT    DEFAULT NULL,
    FOREIGN KEY (project_id)     REFERENCES student_projects(project_id),
    FOREIGN KEY (parent_task_id) REFERENCES tasks(task_id)
);

CREATE TABLE IF NOT EXISTS dependencies (
    dependency_id       INTEGER PRIMARY KEY AUTOINCREMENT,
    predecessor_task_id INTEGER NOT NULL,
    successor_task_id   INTEGER NOT NULL,
    dependency_type     TEXT    NOT NULL DEFAULT 'finish-to-start',
    FOREIGN KEY (predecessor_task_id) REFERENCES tasks(task_id),
    FOREIGN KEY (successor_task_id)   REFERENCES tasks(task_id)
);
```

Code Listing 1. Core table definitions from setup.js

A noteworthy design consideration is that SQLite has no native date type; all dates are stored as `TEXT` in `YYYY-MM-DD` ISO 8601 format [TODO: REFERENCE TS]. This means date parsing is handled on the client side using D3's `d3.timeParse('%Y-%m-%d')` when data is loaded, and dates are serialised back to strings with `d3.timeFormat('%Y-%m-%d')` before being sent in API payloads.

The `comments` table stores comment content, author email, author name, role (constrained to `'student'` or `'supervisor'` via a `CHECK`), a creation timestamp defaulting to `datetime('now')`, and an optional `parent_comment_id` self-reference for threaded replies, supporting **FR22**.

Rather than running a separate migration tool, the server applies idempotent schema changes on startup using `ALTER TABLE ... ADD COLUMN` wrapped in error handlers that silently ignore `duplicate column name` errors. This allowed new columns such as `start_date` and `end_date` on `student_projects` to be added to an existing database without requiring a full reset.

All database interactions occur through REST API endpoints exposed by the Express server. The client communicates with these endpoints via the browser's `fetch` API, issuing `GET` requests to retrieve data and `POST`, `PATCH`, and `DELETE` requests to create, update, and remove records respectively. For example, loading a project's tasks and dependencies on page initialisation is performed in parallel using `Promise.all`:

```
    const [dbTasks, dbDependencies, dbSubtasks, projectBounds] =
   await Promise.all([
    fetchTasks(projectId),
    fetchDependencies(projectId),
    fetchSubtasks(projectId),
    fetchProjectBounds(projectId)
]);
```

Code Listing 2. Parallel data fetch on page load (interactive-chart.js)

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A Use Case Diagrams

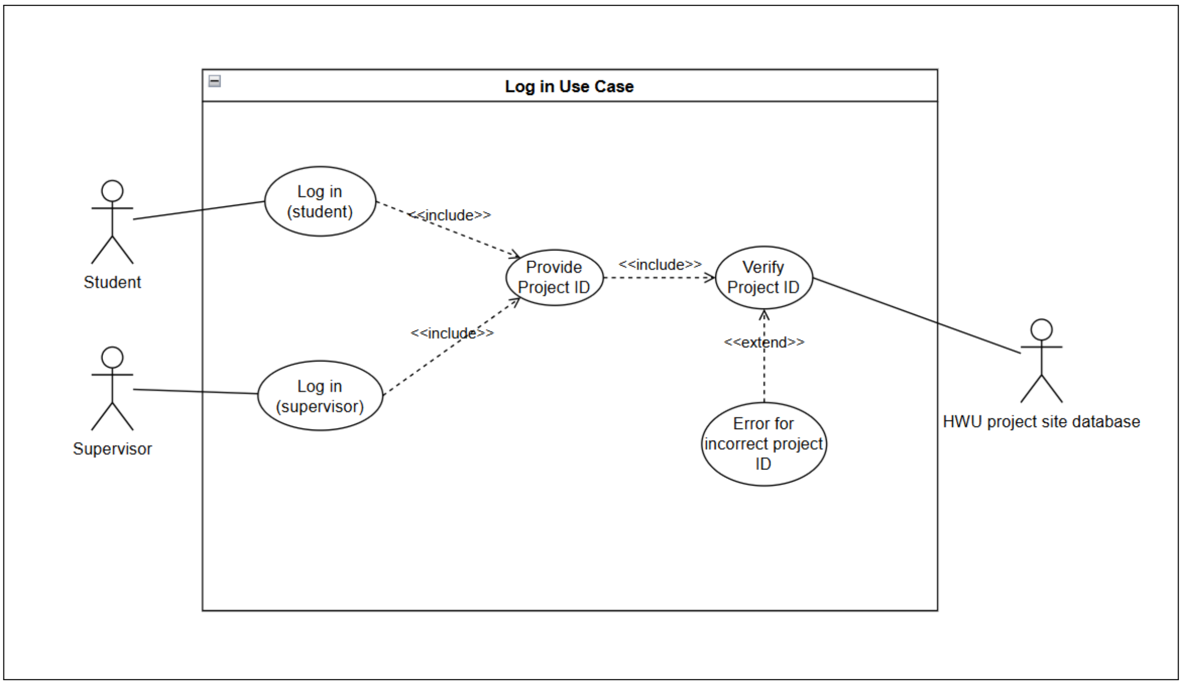

Fig. 1. Log in Use Case Diagram.

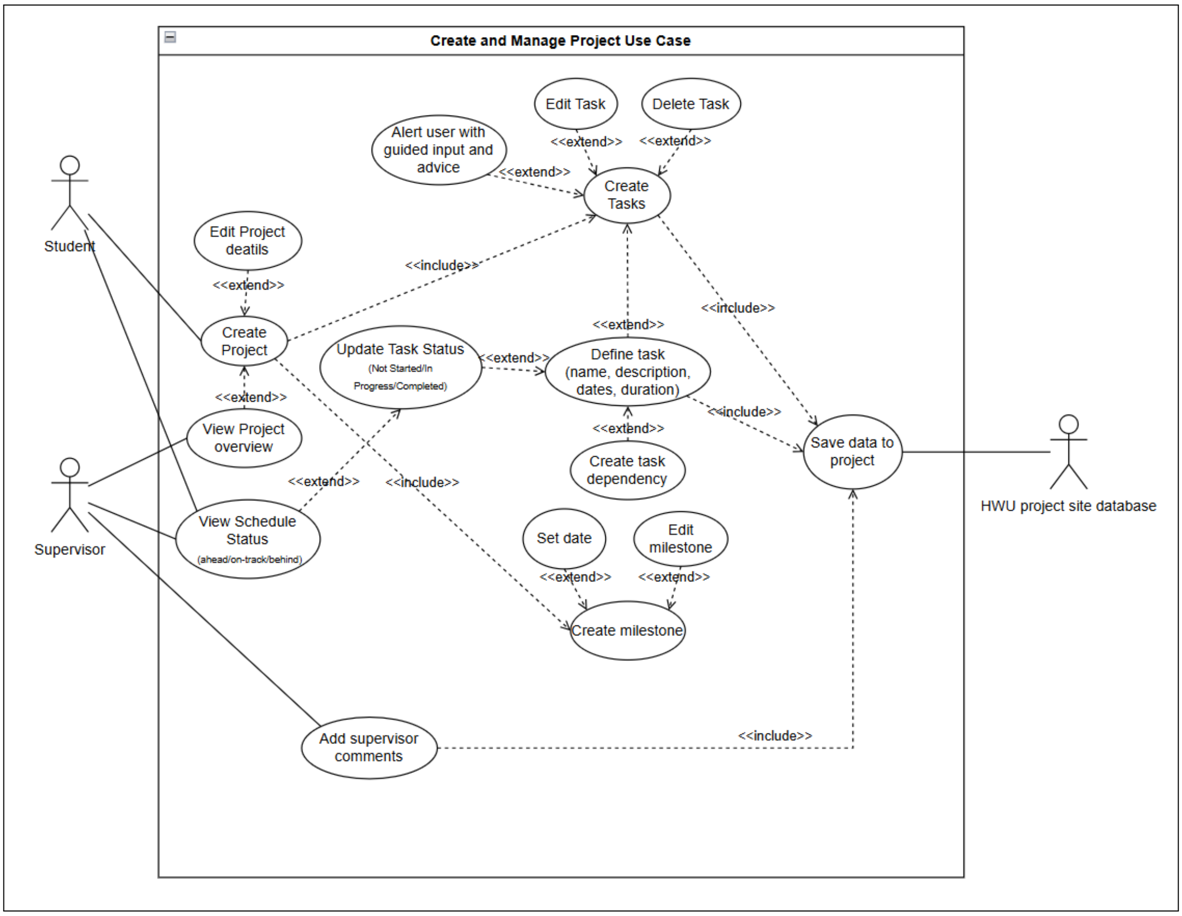

Fig. 2. Create and Manage Project Use Case Diagram.

B Mock Up

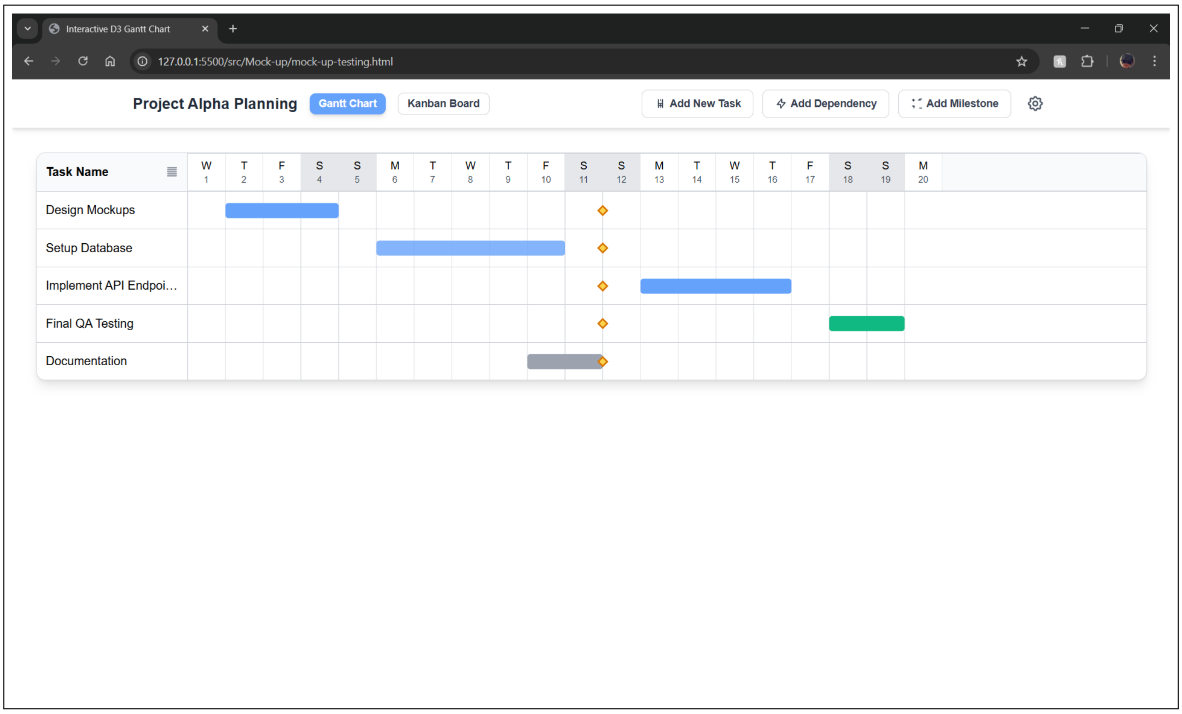

Fig. 3. Rough Mock-ups of the Interactive Gantt Chart Builder Interface.

C Project Management

C.1 Gantt Chart

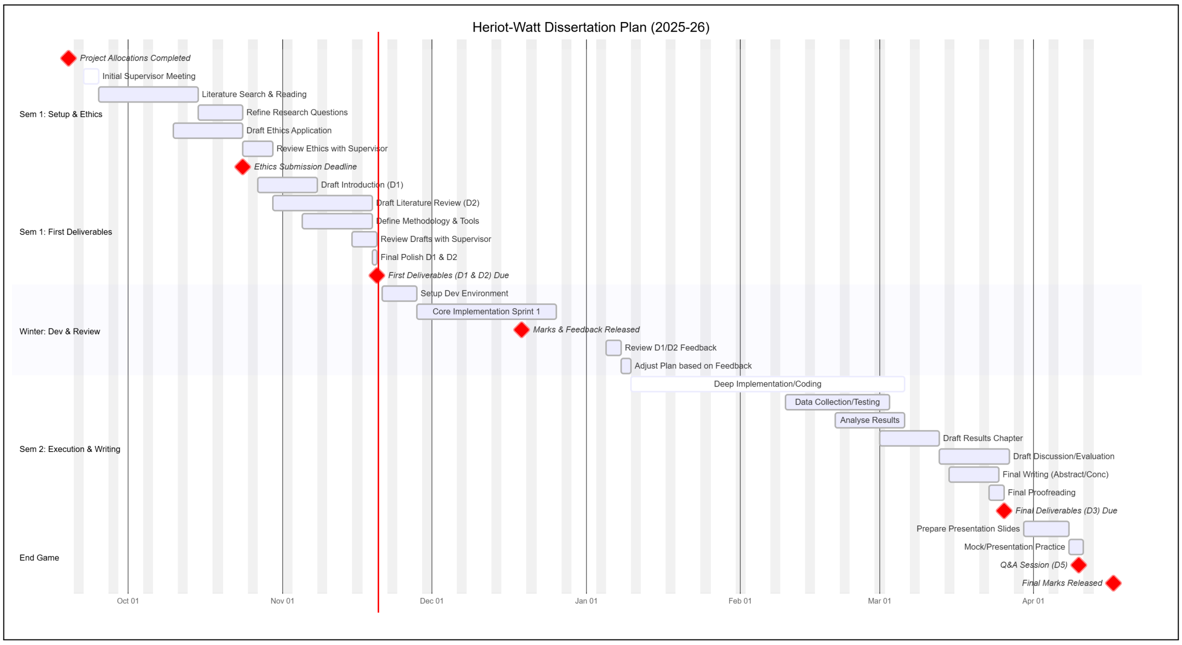

Fig. 4. Gantt Chart for Project Management.

Legend: Red diamonds are milestones; Light blue boxes are tasks; White boxes are supper tasks.

Task / Milestone	Date / Period
Semester 1: Setup & Ethics	
Project Allocations	19 Sep 2025
Initial Supervisor Meeting	22 Sep 2025
Literature Search	25 Sep - 09 Oct
Draft Ethics Application	10 Oct - 20 Oct
Ethics Submission (Deadline)	**24 Oct 2025**
Semester 1: First Deliverables	
Draft Introduction (D1)	27 Oct - 06 Nov
Draft Lit Review (D2)	30 Oct - 13 Nov
Review Drafts	15 Nov - 19 Nov
First Deliverables (D1/D2) (Deadline)	**20 Nov 2025**
Dev Environment Setup	21 Nov 2025
Marks & Feedback	19 Dec 2025
Semester 2: Implementation & Writing	
Review Feedback	05 Jan 2026
Deep Implementation	10 Jan – 20 Feb
Data Collection	10 Feb – 25 Feb
Draft Results Chapter	01 Mar – 10 Mar
Draft Discussion	11 Mar – 20 Mar
Final Polish (D3)	21 Mar – 25 Mar
Final Dissertation (D3) (Deadline)	**26 Mar 2026**
End Game	
Viva Preparation	30 Mar – 09 Apr
Q&A Session (D5)	**10 Apr 2026**
Final Marks Released	17 Apr 2026
Expo (Optional) (D6)	TBC

Table 1. Project Schedule

C.2 Risks

The development of the Interactive Gantt Chart Builder involves several identified risks, each assessed for likelihood and impact, with appropriate mitigation strategies.

Risk	**Level**	**Mitigation Strategy**
Technical Implementation	Medium	Preliminary D3.js mock-ups validate feasibility. Iterative development with regular testing will identify performance issues early. Established technologies reduce implementation uncertainty.
Integration with HWU System	Low-Medium	Early stakeholder consultation established clear technical constraints. System inherits existing authentication infrastructure. Regular communication with developers will address emerging concerns.
Timeline Delays	Medium	Project schedule incorporates buffer time. MoSCoW allows scope adjustment whilst ensuring core functionality delivery. Regular supervisor meetings provide early warning of issues.
Usability Evaluation Recruitment	Low	Participants recruited from honours cohort with adequate notice and flexible scheduling. Expert review can supplement user testing if needed.
Scope Creep	Medium	Requirements clearly documented and prioritised. Should and Could have requirements provide clear boundaries. Regular stakeholder communication manages expectations.
Data Security	Low	All data stored on secure HWU systems. Evaluation uses fictitious data. System follows established web security practices including input validation and secure database access.

prioritisation

Table 2. Project Risk Assessment

Overall, the project's risk profile is manageable, with most risks rated as low to medium severity. The use of proven technologies, early stakeholder engagement, and flexible requirements prioritisation provide a robust foundation for successful project completion.

C.3 PLES Issues

The development and evaluation of the Interactive Gantt Chart Builder addresses Professional, Legal, Ethical, and Social (PLES) considerations as follows, adhering to the British Computer Society (BCS) Code of Conduct [7].

Professional: The system implementation prioritises professional competence and integrity [7]. Development follows established software engineering principles and secure coding standards (e.g., OWASP guidelines [34]) to prevent vulnerabilities. The code is documented to industry standards to ensure maintainability for future HWU developers.

Legal: The project strictly complies with the UK Data Protection Act 2018 [21] and the GDPR [12]. All participant data is anonymised and stored on secure University servers. Furthermore, the project adheres to Intellectual Property (IP) rights; the software is an original creation, and any third-party libraries used are compliant with their respective open-source licences (e.g., MIT, Apache), ensuring no copyright infringement.

Ethical: In line with the University's ethical approval process, all evaluation participants will provide informed consent via information sheets, retaining the right to withdraw at any time without penalty. To mitigate harm, the system uses fictitious project data during testing, ensuring no student's actual academic work is at risk during the beta phase.

Social: The tool promotes digital inclusion by addressing the needs of students with varying levels of project management experience. To ensure equitable access, the interface aims to meet WCAG 2.1 AA accessibility standards [26], complying with the Equality Act 2010 [20], ensuring that students with visual or motor impairments can utilise the educational tool effectively.

D Generative AI

Generative artificial intelligence tools were used during the preparation of this dissertation to assist with literature search and reference discovery, and to support writing quality through spelling, grammar, punctuation, and sentence structure improvements. Generative AI was also used in troubleshooting isues with text editors such as LaTex in the creation of this document. All references identified through AI assistance were systematically verified by hand to ensure legitimacy, relevance, and that cited content accurately supported the claims made. Generative AI functioned solely as a supportive tool rather than a generator of primary content or ideas.